

The Muffin Problem

Spring 2024 ARML Power Contest

There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth. Be sure to read the problems carefully and answer the question that is being asked!

Muffy Baker is a student in her school's math club, which has a total of five students (including her) in it. She baked some muffins for the club meeting. She baked five muffins, one for each student. Unfortunately, two of the students couldn't come at the last minute. To divide the five muffins among the remaining three students fairly, each student should get $5/3$ of a muffin. So Muffy gave each person one whole muffin, and cut $1/3$ out of each of the remaining two muffins. She took the two $1/3$ -sized pieces, and gave the other two students $2/3$ -sized pieces. The muffins were thus divided according to the table below:

	m_1	m_2	m_3	m_4	m_5
s_1	1			$\frac{1}{3}$	$\frac{1}{3}$
s_2		1		$\frac{2}{3}$	
s_3			1		$\frac{2}{3}$

Note that each student receives a total of $5/3$ muffins. Being curious, the students note that the smallest piece anyone receives is $1/3$ of a muffin. They wonder if they could do better; could they have divided the muffins so that the smallest piece anyone would get would be larger than $1/3$?

It turns out the answer, in this case, is YES! By playing around, the students discover the following way of dividing the muffins:

	m_1	m_2	m_3	m_4	m_5
s_1	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{5}{12}$	
s_2	$\frac{7}{12}$	$\frac{7}{12}$			$\frac{1}{2}$
s_3			$\frac{7}{12}$	$\frac{7}{12}$	$\frac{1}{2}$

The math club decides to investigate further. They create a function $f(m, s)$ for positive integers m and s whose value is the “largest smallest piece.” That is, $f(m, s) = p$ where p is the largest possible number so that m muffins can be equally shared by s students, and no student has to receive a piece smaller than p . So far, the students have determined that $f(5, 3) \geq 5/12$.

The Problems

1. Explain why $f(m, 1) = 1$.

[1 pt.]

2. Compute, with explanation, $f(1, s)$. [2 pts.]
3. Compute, with explanation $f(ks, s)$ where k is any positive integer. [2 pts.]
4. Explain why, if m is *not* a multiple of s as in the previous problem, $f(m, s) \leq 1/2$. [2 pts.]
5. Any way of dividing m muffins fairly among s students leads to *muffin division tables* such as those created by the students to display how they divided 5 muffins among 3 students. Explain why the numbers in each column of a muffin division table must add to 1. [1 pt.]
6. Compute, with explanation, the sum of the numbers in any row of a muffin division table. [2 pts.]

The next few problems will show that $f(5, 3) = 5/12$. That is, there is no way to cut up five muffins so that all students can receive $5/3$ of a muffin and the smallest piece anyone must receive is larger than $5/12$. Since 5 is not a multiple of 3, from problem 4 there must be at least one piece that is no larger than $1/2$ in size. Thus, if any muffin in any division remains uncut (i.e. one student received a whole muffin as well as some other pieces) then instead cut that muffin into two pieces of size $1/2$ and give both to that student. Since $f(5, 3) \leq 1/2$ this cut doesn't change the size of the smallest piece anyone will receive. Thus, it may be assumed that in the division of the muffins that guarantees the largest possible smallest piece, all muffins are cut into at least two pieces.

7. Explain why, to make the largest possible smallest piece, no muffin should be cut into more than two pieces. [2 pts.]
8. Since, by the previous problem, all muffins should be cut into two pieces, there are ten pieces of muffins. Use the pigeon-hole principle to show that someone must have a piece that is no larger than $5/12$. [2 pts.]

Hence, $f(m, s) \leq 5/12$. Since a method of dividing the muffins was found where the smallest piece was $5/12$ this completes the proof that $f(5, 3) = 5/12$. The next few problems will show that $f(3, 5) = 1/4$.

9. Create a muffin division table that divides 3 muffins among 5 students, where no piece of any muffin has size less than $1/4$. [2 pts.]
10. If someone gets a single piece of size $3/5$, explain why someone else must get a piece that is smaller than $1/4$. (Hint: what happens to the other $2/5$ of the muffin from which the $3/5$ piece was taken?) [2 pts.]
11. If there is a division where no one gets a piece smaller than $1/4$, explain why no one gets three or more pieces. [1 pt.]
12. So in any division where the smallest piece is $1/4$ everyone gets exactly two pieces. So the muffins are divided into a total of ten pieces. Complete the proof that $f(3, 5) = 1/4$. [2 pts.]

Note that $f(5, 3) = 5/12$ and $f(3, 5) = 1/4$. Did you notice that $\frac{5}{12} = \frac{5}{3} \cdot \frac{1}{4}$? That is, $f(5, 3) = \frac{5}{3}f(3, 5)$. This is no accident!

13. Use muffin division tables to prove that $f(m, s) = \frac{m}{s}f(s, m)$ for all positive integers m and s . [3 pts.]

The technique used to establish the value for $f(5, 3)$ may be generalized to exhibit an upper bound for $f(m, s)$ called the *floor-ceiling upper bound*. The next few problems will establish that, if m is not a multiple of s , then

$$f(m, s) \leq FC(m, s) = \max \left\{ \frac{1}{3}, \min \left\{ \frac{m}{s \lceil 2m/s \rceil}, 1 - \frac{m}{s \lfloor 2m/s \rfloor} \right\} \right\}.$$

As before, using problem 4 it may be assumed that every muffin is cut into at least two pieces.

14. Explain why if there is some muffin that is cut into three or more pieces the upper bound is satisfied. [1 pt.]
15. On the other hand, if every muffin is cut into exactly two pieces, there are $2m$ pieces of muffin. Prove that $FC(m, s)$ is an upper bound for the smallest-sized piece that someone must receive. [3 pts.]

Since the above cases cover all possibilities, the floor-ceiling upper bound for $f(m, s)$ has been proven.

16. Prove that the floor-ceiling upper bound is tight when there are three students and $m > 1$ is not divisible by three. (Note: $FC(1, 3)$ is undefined, but it is already known that $f(1, 3) = 1/3$. Similarly, it is already known that $f(3k, 3) = 1$.)
- (a) Compute and simplify $FC(3k + 1, 3)$ for $k \geq 1$. [1 pt.]
 - (b) Compute and simplify $FC(3k + 2, 3)$ for $k \geq 0$. [1 pt.]
 - (c) For $k \geq 1$ describe a division of $3k + 1$ muffins among three students that achieves the upper bound $FC(3k + 1, 3)$ of the smallest required piece. [2 pts.]
 - (d) For $k \geq 0$ describe a division of $3k + 2$ muffins among three students that achieves the upper bound $FC(3k + 2, 3)$ of the smallest required piece. [2 pts.]

The floor-ceiling upper bound also gives the correct answers for $s = 4$. It gives the correct answers for $s = 5$ *except* when $m = 11$.

17. Note that $FC(11, 5) = 11/25$, but the correct value of $f(11, 5)$ is actually $13/30 < 11/25$. Find a division of 11 muffins among 5 students that achieves this value of $13/30$ for the smallest piece. [2 pts.]
18. For a final challenge, prove that $f(24, 25) = 8/25$. [4 pts.]

Contest ends here.

No contest material on this page.

Much more is known about the function $f(m, s)$. For examples, it is simple to see that $f(km, ks) \geq f(m, s)$ because any strategy for splitting up m muffins among s students could just be repeated k times to split up km muffins for ks students, and the smallest piece required wouldn't get any smaller! It is reasonable to think, though, that the extra flexibility with more muffins and more students might allow the smallest piece to increase in size. But this is not the case! It is not simple to prove, but $f(km, ks) = f(m, s)$.

Together with the fact proven in the contest that $f(m, s) = \frac{m}{s}f(s, m)$ this shows that $f(m, s)$ can be computed for all values of m and s just by considering those that are relatively prime and for which $m > s$. In fact, the problem can be turned into a 1-variable problem by considering the function $g(x) = f(m, s)$ for $x = m/s$. This function is very badly behaved, and doesn't exhibit much regularity in its graph. Check out the graphs [here](#) for some plots of $g(x)$ over several different intervals.

The muffin problem is actually a fairly new mathematical puzzle, created in 2010 by a gentleman named Alan Frank. It was shared in some local puzzle forums for several years before showing up in a pamphlet of fun problems compiled by Nancy Blachman for the Julia Robinson Mathematics Festival. Some of these pamphlets found their way to the 2016 Gathering 4 Gardner conference. (Martin Gardner was one of the foremost puzzlersmiths and mathematical communicators of the 20th century. He wrote the mathematical games column for *Scientific American* for many years before it was taken over by *Gödel, Escher, Bach* author Douglas Hofstadter.) William Gasarch picked up a copy of the pamphlet and found the problem fun, and had his graduate students start to write programs to solve various cases.

In 2018, Scott Huddleston wrote a computer program that solves the muffin problem very quickly (linear time complexity!) but no one had a proof that it worked correctly in all cases. That would have to wait for 2020 when Richard Chatwin solved an isomorphic problem and proved the solution worked. His paper is available on the arXiv at arxiv.org/pdf/1907.08726.pdf. (It is not light reading!). Gasarch and his students published a book about the topic, so if you want to know everything there is to know about the muffin problem, here is a link to the [book on Amazon](#). (ARML receives no consideration for purchases of this book.)

The problem is recent enough, in fact, that the [On-Line Encyclopedia of Integer Sequences](#) doesn't know very much about it! The only sequence in the encyclopedia as of the end of 2023 is [sequence A298783](#) which tells how often the floor-ceiling upper bound fails to give the correct answer.

If you've enjoyed this contest, treat yourself to some muffins! Hopefully you can cut them so that no one gets only tiny crumbs!